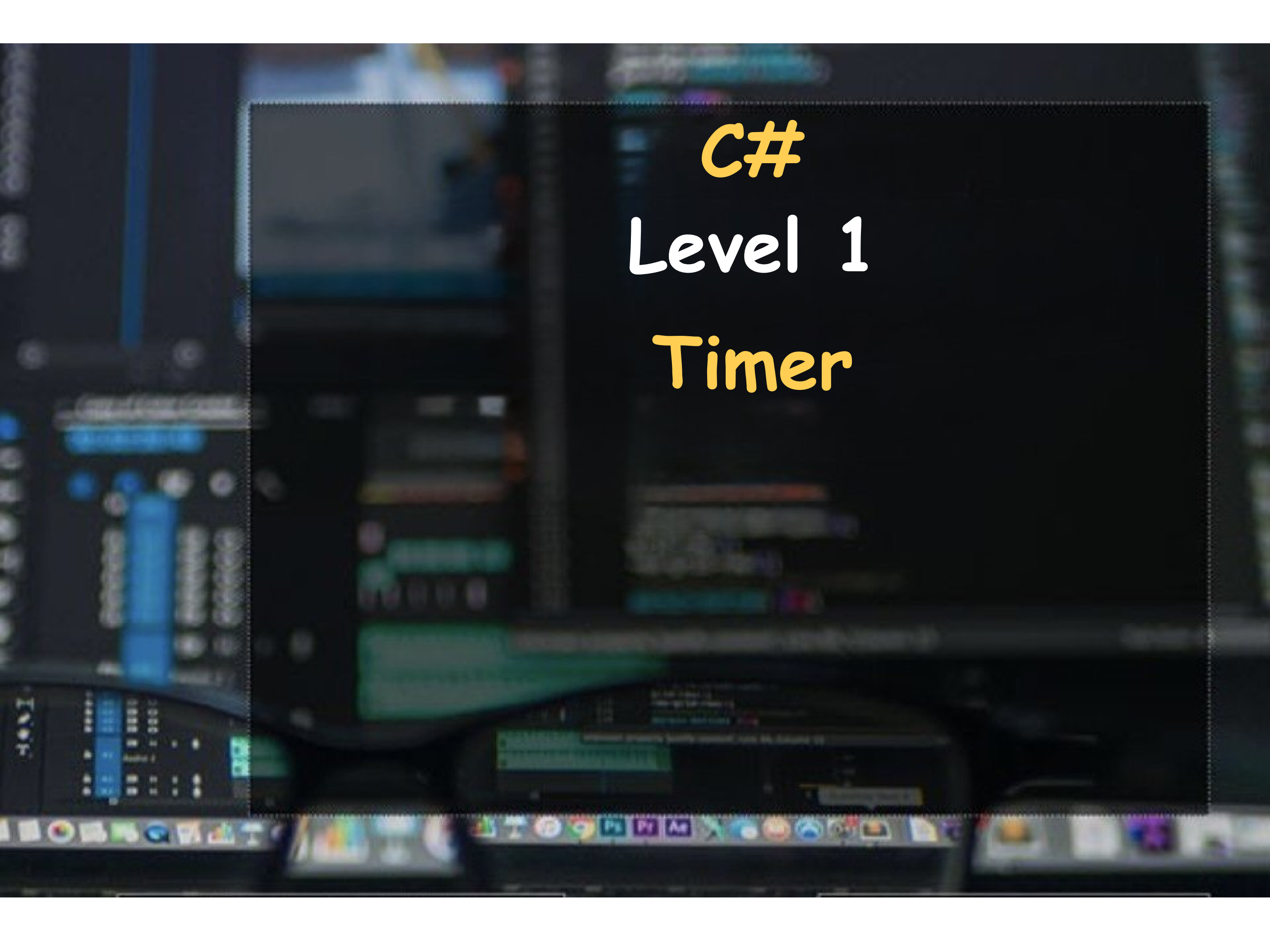
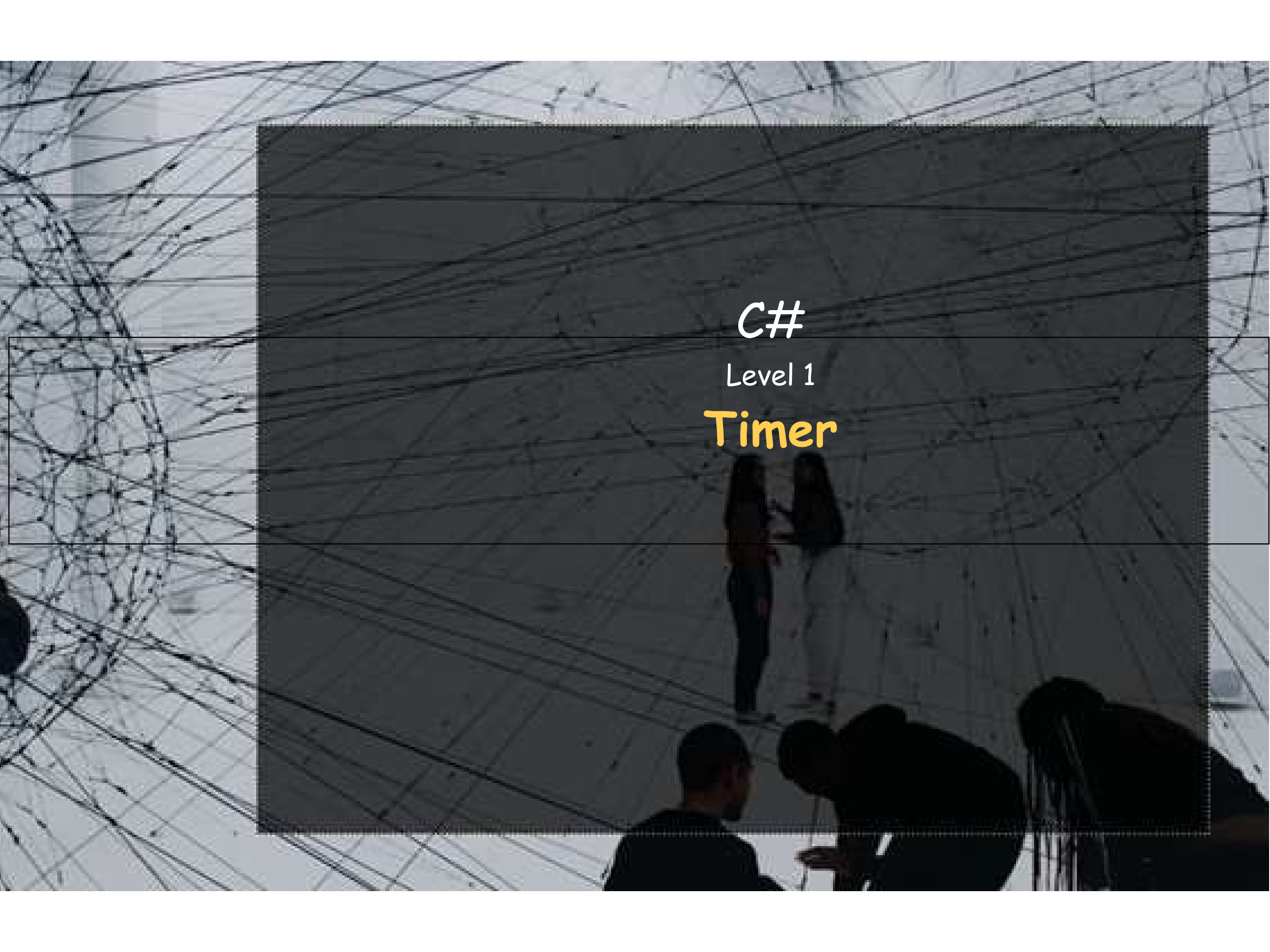




C# Level 1 Timer



C#

Level 1

Timer

Definition

Timer

 Windows Forms component that raises the Tick event repeatedly at a specified interval.

 In this example, it is used to create a simple counter.

Explanation

Counter

-  Counter stores the current count and starts at 0.

-  Counter++ increases the value by 1 on every Tick event.

Display

-  label1.Text displays the current Counter value.

-  ToString() converts the integer into text.

Timer Control

-  timer1.Enabled = true starts the Timer.

-  timer1.Enabled = false pauses the Timer.

Procedure

1. Initialize Counter to 0.
2. Start the Timer with Enabled = true.
3. Increase Counter on every Tick event.
4. Display the updated Counter in label1.
5. Pause the Timer by setting Enabled = false.
6. Resume it by setting Enabled = true.
7. Reset Counter to 0 when required.

💡 Example

```
private void timer1_Tick(object sender, EventArgs e)
{
    Counter++;
    label1.Text = Counter.ToString();
}
```

➡ Result

- Button 1 → Starts the Timer and counter.
- Button 2 → Pauses the Timer and keeps the current value.
- Button 3 → Resumes the Timer from the current value.
- Button 4 → Stops the Timer and resets the counter to 0.

Note

- ◆ Interval specifies the time between Tick events in milliseconds.
- ◆ For example, Interval = 1000 means approximately 1 second.
- ◆ Tick occurs whenever the specified interval elapses.



Comparison



Pause

- ✓ Stops Tick events.
- ✓ Keeps the current Counter value.
- ✓ Can continue later from the same value.



Reset

- ✓ Stops the Timer.
- ✓ Sets Counter to 0.
- ✓ The next start begins from zero.

⊘ Warning

- ◆ Enabled = false pauses the Timer but does not reset Counter.
- ◆ Resetting requires setting Counter = 0.
- ◆ Windows Forms Timer runs Tick on the UI thread, so long-running code can make the interface unresponsive.

✦ Summary

- ✓ Timer executes code repeatedly through the Tick event.
- ✓ Interval controls the time between Tick events.
- ✓ Enabled starts or pauses the Timer.
- ✓ Counter++ increases the counter.
- ✓ label1.Text displays the counter.
- ✓ Pause keeps the current value, while Reset sets it back to 0.

Thank you 😊